



SYENSQO
ADVANCING HUMANITY

Macro to Micro:

Processing, Properties, and
Simulation Techniques

James Hicks, Elias Irick

MD&M West, February 2026



Agenda

1. Introduction to Syensqo

- Who We Are
- Materials for Demanding Applications

2. Micro Injection Molding at Syensqo

- Overview & Motivation
- Machine
- Tooling

3. Processing (vs Macro)

- Nozzle
- Temperatures (Screw, Barrel, Mold)
- Speeds & Pressures

4. Properties (vs Macro)

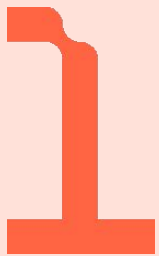
- Fiber Orientation
- Crystallinity
- Moisture Uptake Rate
- Mechanical Strength

5. Simulation (vs Macro)

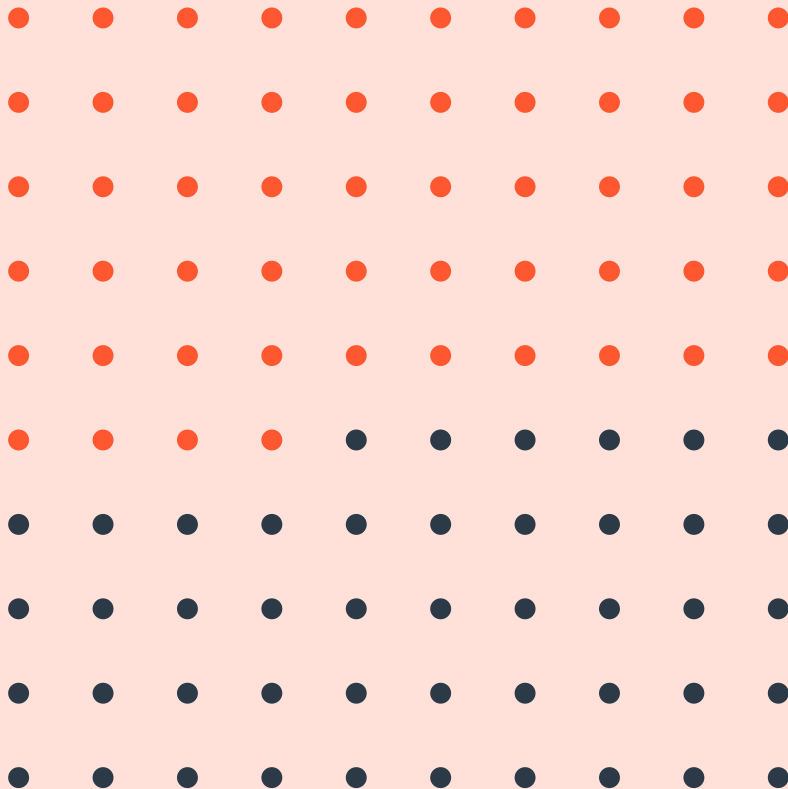
- Back end (material cards)
- Front end (consumer)

6. Conclusion

- Recap
- Ongoing Efforts & Future Work



Introduction to Syensqo

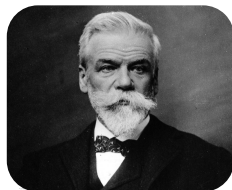


160 Years of Change: from Solvay to Syensqo



1863

Ernest Solvay invents a new process for producing soda ash



Solvay builds first soda ash industrial plant in Couillet site, Belgium

1866

1911

Ernest Solvay launches 1st Solvay Science Conference

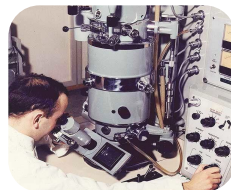


Solvay embraces the plastic revolution

1950

1958

Solvay's first hydrogen peroxide (H₂O₂) plant in Jemeppe-sur-Sambre, Belgium



With pharma, Solvay becomes a **three segments** company

1980

2000

Solvay acquires specialties



Solvay acquires Cytac

2015

2022

Launch of breakthrough e.Solvay process



2023
Launch of Syensqo

2023
New Solvay



Syensqo is a Market Leader in Materials & Consumers



MATERIALS

Specialty Polymers

High-performance polymers; leading position in thermoplastic composites

Battery Materials, Thermoplastic composites, Green Hydrogen, Renewable materials & Biotechnology

Composites

Materials for civil aerospace

HPP glues

Materials for defense

CONSUMER & RESOURCES

Novecare

Specialty surfactants and polymers

Aroma

Flavors & Fragrances; Natural Vanillin

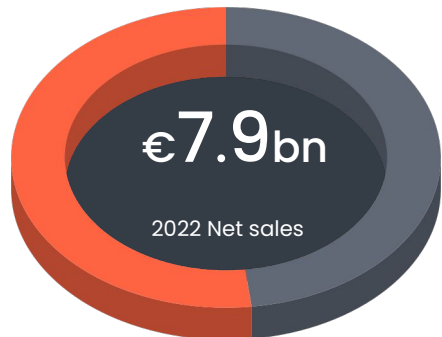
Technology solutions

Mining reagents

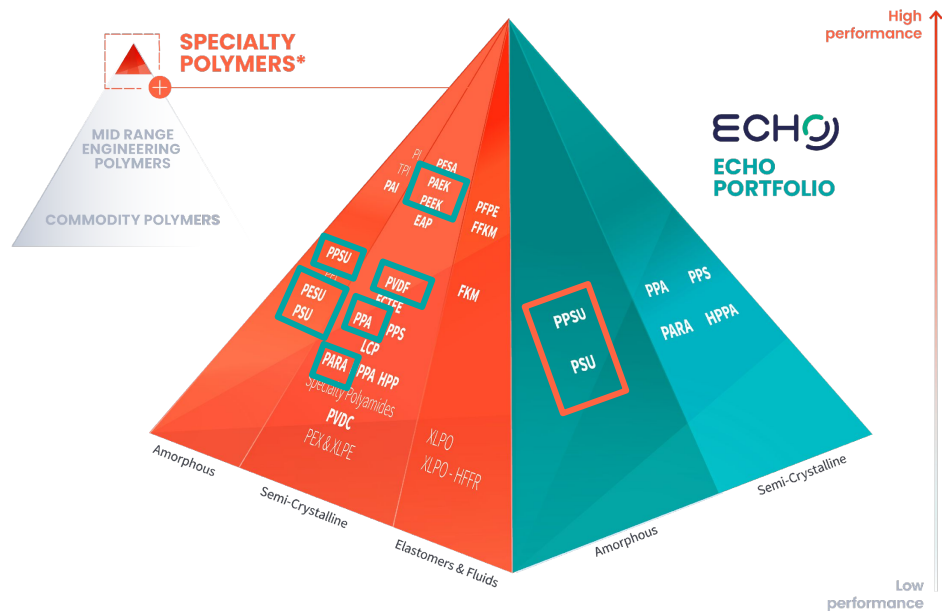
Oil & gas Solutions

Biocides for recycled water

52%
Materials



Syensqo Specialty Polymers Overview



High-Performance Medical Grade Plastics

Limited exposure < 24 hours
ISO 10993 Biocompatibility

- KetaSpire® PEEK
- AvaSpire® PAEK
- Ixef® PARA HC and GS
- Radel® PPSU
- Udel® PSU
- Veradel® HC PESU
- Solef® PVDF
- Amodel® PPA*

*In development, sampling available



Lightweighting



Fire resistance



Mechanical performance



Resistance to harsh elements



Broad range of temperatures



Surface properties

Excellence in Medical for 60 years

Wide range of medical-grade thermoplastics for medical devices, implants, and bioprocessing systems



Udel® PSU

1965

Replaced cellulose acetate in blood filtering membranes, drastically improving patient experience



Ixef® PARA

1988

Single-use specialty polymer utilized in surgical instrumentation



Radel® PPSU

1990

High performance specialty polymer effective in a wide range of medical applications



KetaSpire® PEEK
AvaSpire® PAEK

2006

High performance polymers renowned for their exceptional resilience in the most demanding medical environments



Solviva® Implantable
PEEK, PPSU, PSU

2008

Orthopedic & Active implant biomaterials:
Zeniva® PEEK
Eviva® PSU
Veriva® PPSU

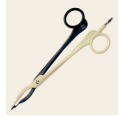
Manufactured in
ISO 13485 and cGMP
compliant facility in
Georgia, USA



ECHO®
Portfolio

2024

High-quality, sustainable alternatives to reduce the industry's environmental impact



Amodel® PPA
Solef® PVDF

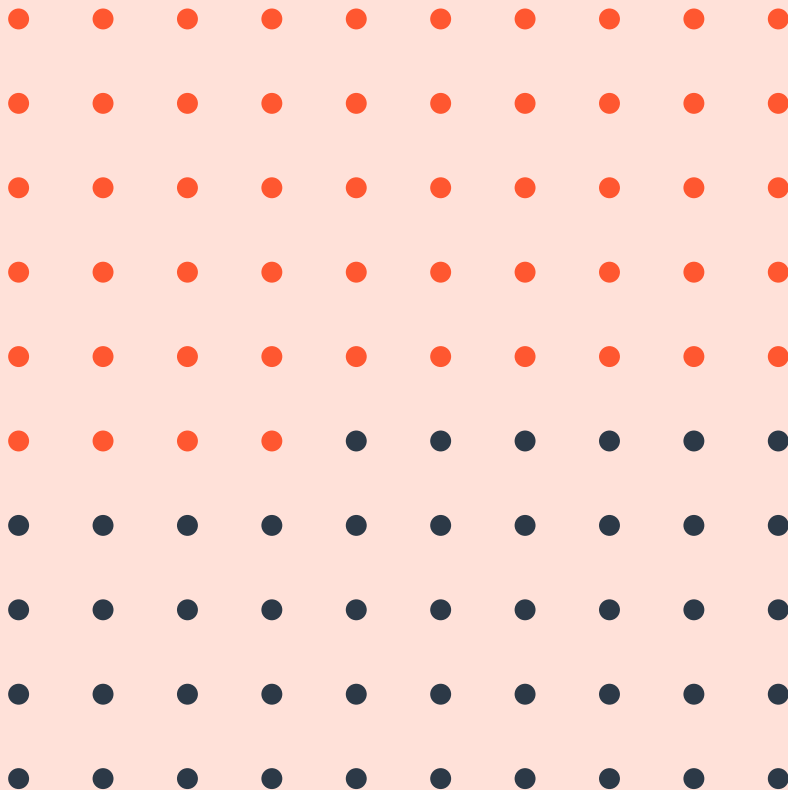
2025

Amodel® PPA
Increased temperature resistance & electrical properties

Solef® PVDF
Extreme inertness in harsh environments



Micro Injection Molding



Micro Injection Molding Overview



Legacy internal expertise in
“macro” injection molding:

- Material characterization
- Simulation
- Prototyping
- Customer troubleshooting

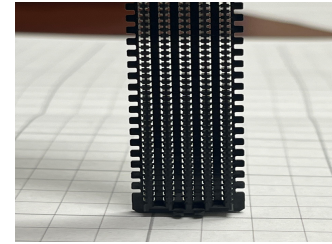
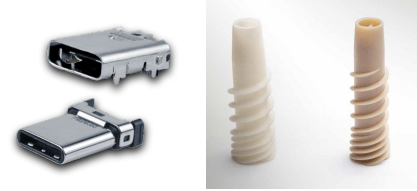
Customers in several markets use our materials in **micro** injection molding applications:

Healthcare (implantables, screws)
Electronics (connectors, camera components)

Macro to Micro Injection Molding Advantages:

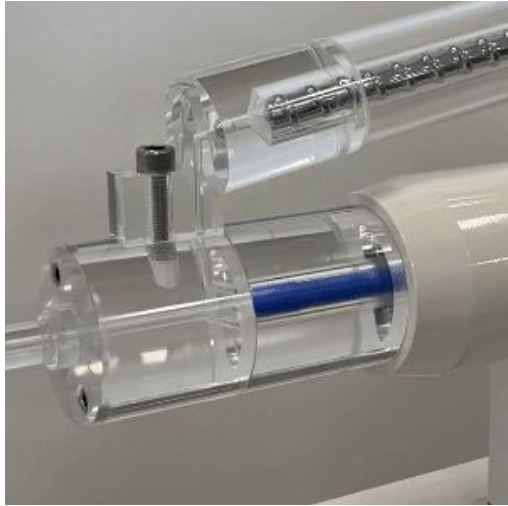
- Precision/accuracy
- Rapid cycling
- Low cavitation = material savings

Syensqo began a global coordinated effort in 2024
to build competency and expertise in micromolding

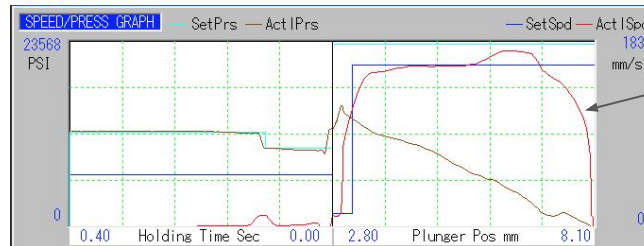


Syensqo Micromolding Machine

- Long relationships with many vendors who make micro molding machines:
- Acquired **Sodick GL30-LP** with optional V-Connect software
- Selection criteria was based on our R&D use case



Demo unit showing the decoupled injection and plasticization mechanisms.



 Sumitomo

ARBURG

NISSEI



Sodick

Repeatable **control** and **precision** even at extremely fast (**<60ms!**) injection times.

Raw pressure and plunger position **data can be exported** for simulation and study.

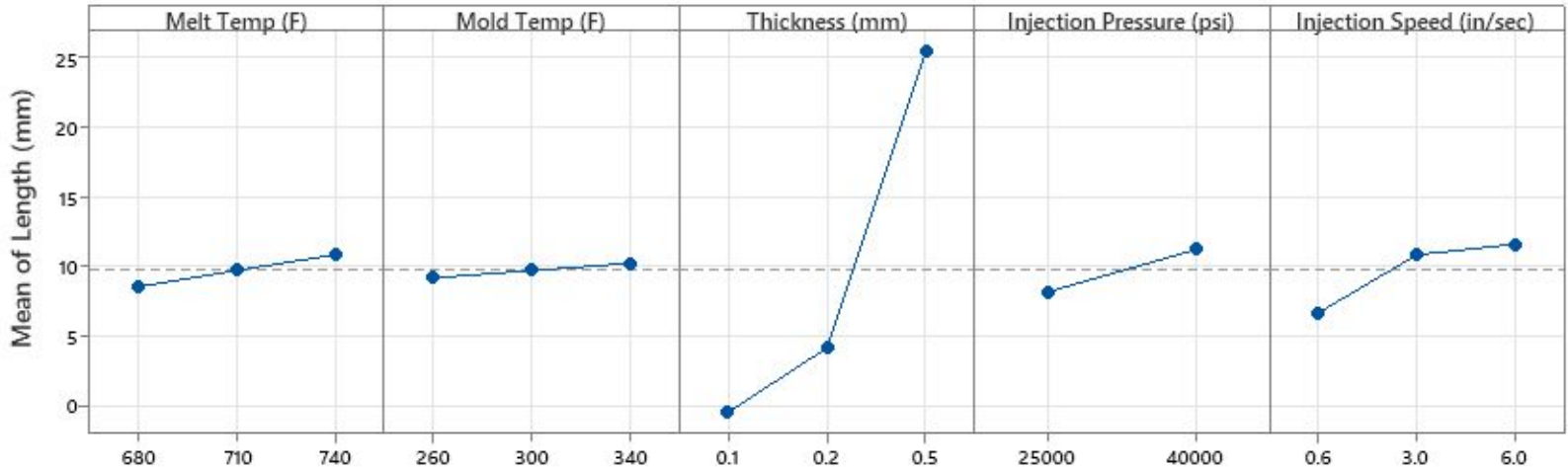
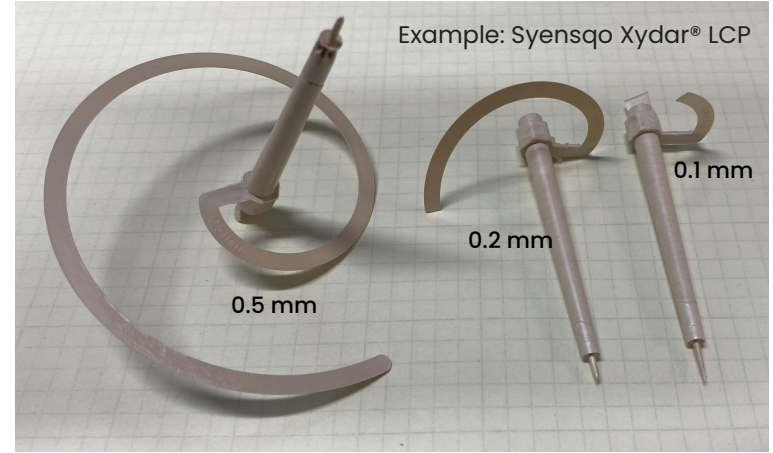
Syensqo Micromolding Tooling: **Spiral**

250 mm maximum flow length

Shut off at sprue to **select depth**:

- 0.5 mm
- 0.2 mm
- 0.1 mm

Pressure transducers can be installed behind ejector pins in 5 locations.



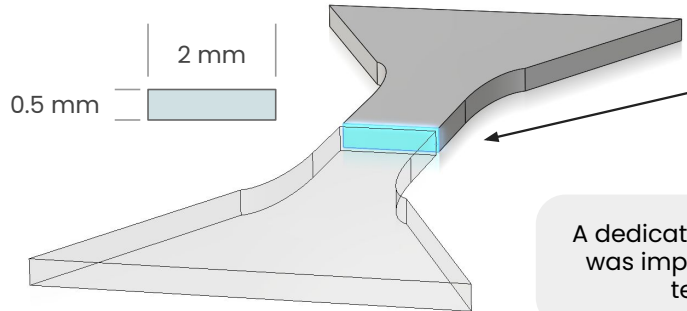
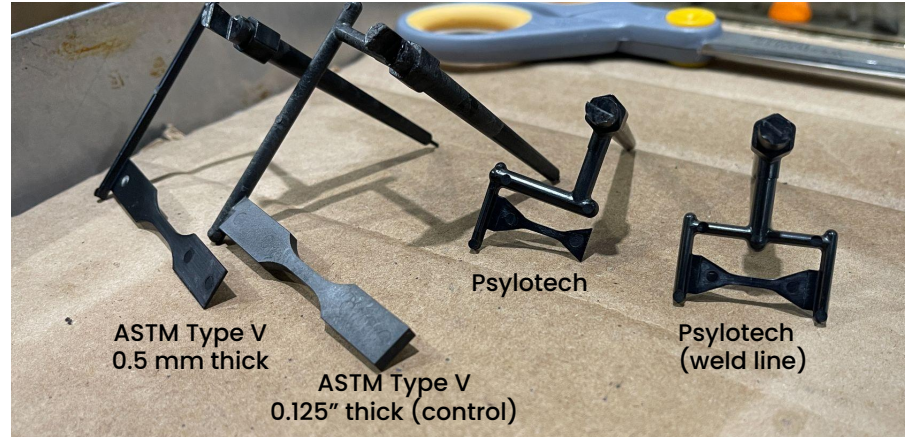
Syensqo Micromolding Tooling: **Tensile Bar Family**

Family tool with selectable cavities

- Designed to benchmark material properties for micromolding applications
- With and without weld line

Nominal thickness = **0.5 mm**

(± 0.001 " is 5% of the thickness)



1 mm² gage area
(1 MPa = 1 N/mm²)

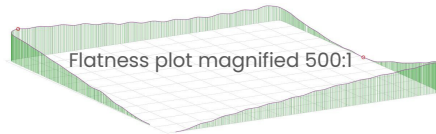
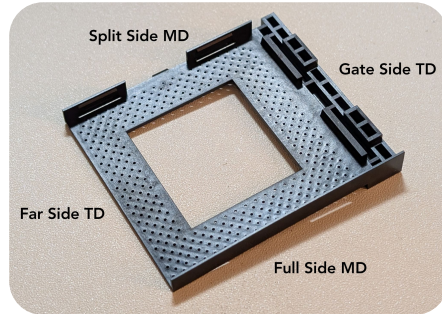
A dedicated mechanical **test method** was implemented for this Psylotech tensile bar specimen.



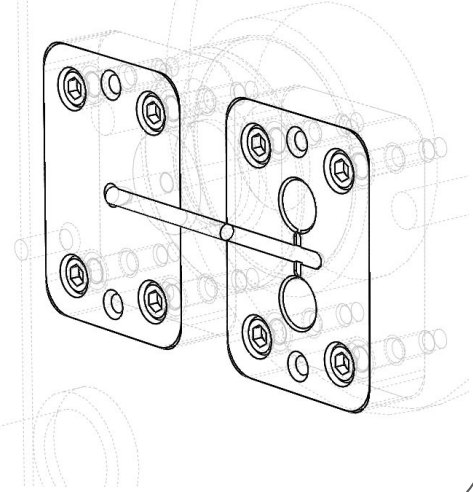
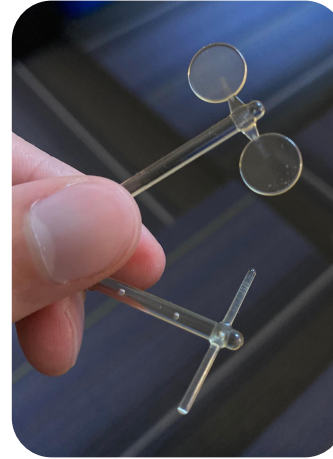
Syensqo Micromolding Tooling: **Others**

PGA (Pin Grid Array):

Thin walled CPU **connector** for flow / warpage analysis.



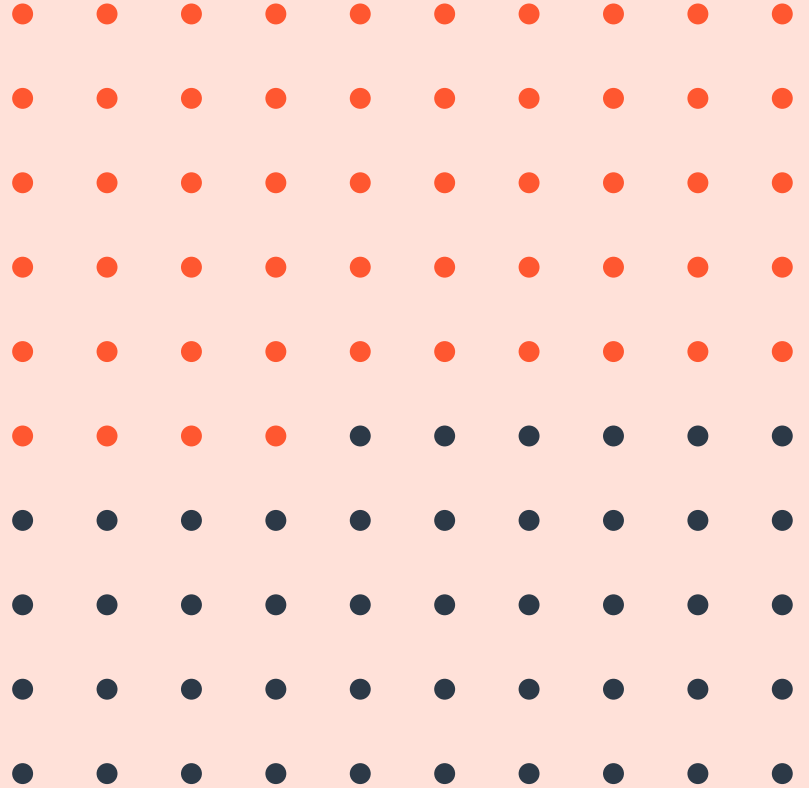
Pocketed mold base with **mini discs and rods** for testing implantable material grades (ISO 10993)



We have more customer-facing molds (confidential designs) to develop materials for specific applications.



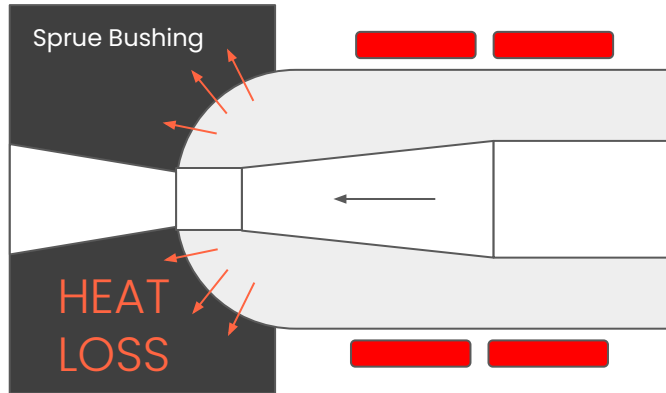
Micromolding Processing



Micro Injection Molding - Nozzle

General Purpose Nozzle

Heater Bands

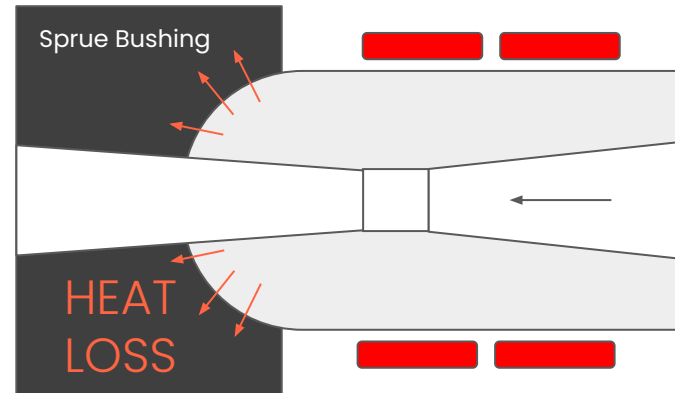


Sprue break location inconsistency

Problematic when parts are the same order of magnitude as a cold slug

Reverse Taper Nozzle

Heater Bands



Consistent fracture location

Shot-to-shot repeatability

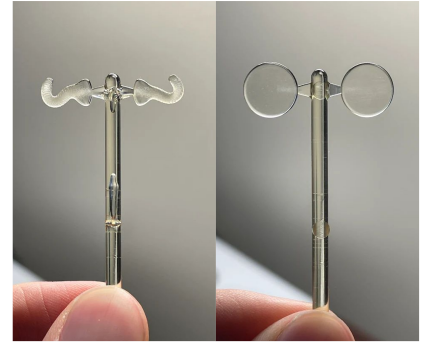
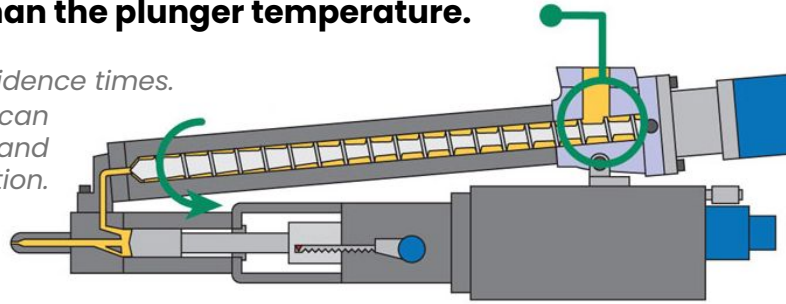
Micro Injection Molding Processing – Temperatures

For screw-over-plunger machines:

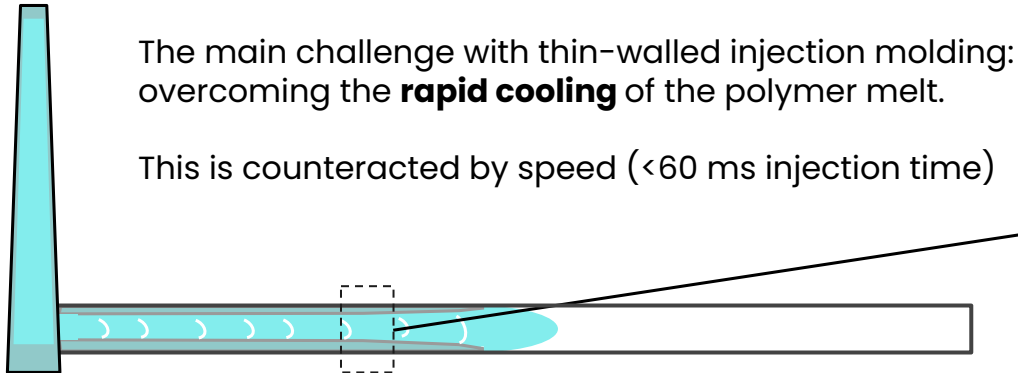
Set the screw $\sim 10^\circ$ lower than the plunger temperature.

Small shot sizes mean long residence times.

*Lowering the screw temps can
improve material conveyance and
reduce degradation.*

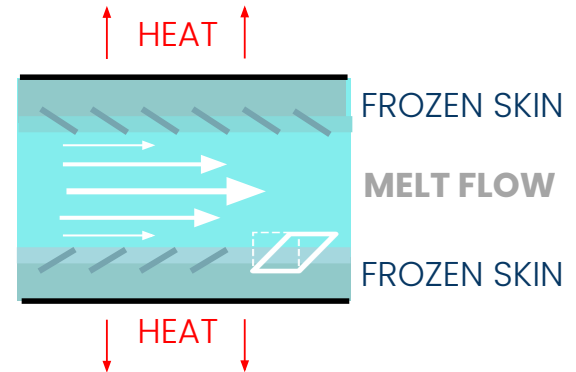


Short shot with visible jetting due to high injection speeds



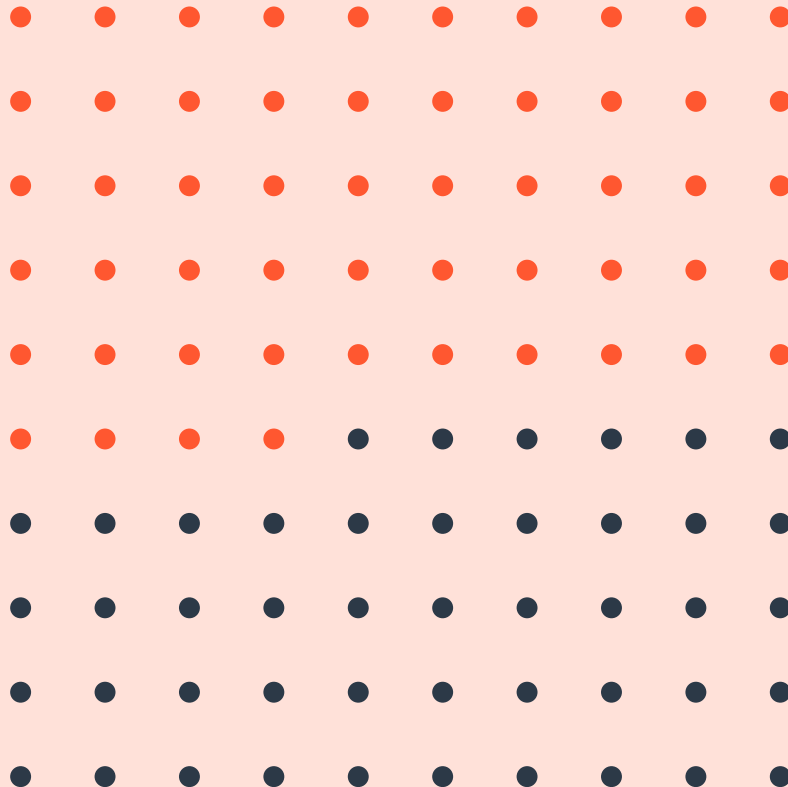
The main challenge with thin-walled injection molding:
overcoming the **rapid cooling** of the polymer melt.

This is counteracted by speed (< 60 ms injection time)



4

Micromolding Properties



Macro to Micro Properties: **Fiber Orientation**

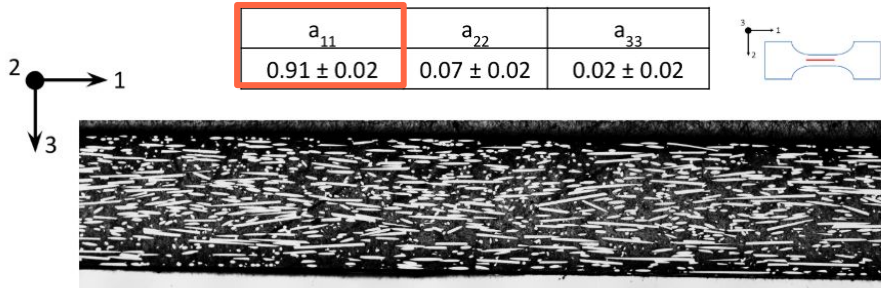
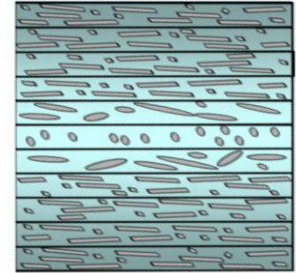
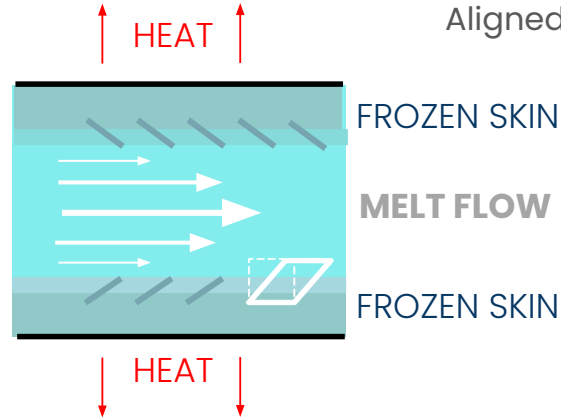
Thin parts show **more fiber alignment**

Typical Digimat fiber alignment tensor: 0.80–0.85
(in the direction of flow)

Thin (0.5mm) tensile bar has a value of **0.90**

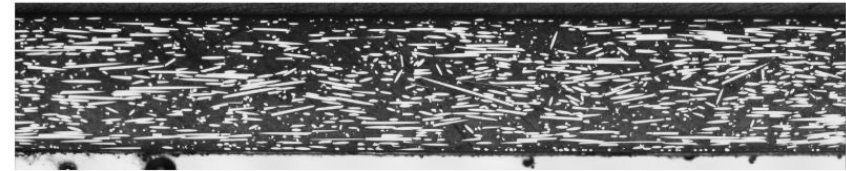
No skin-core effect **regardless of injection speed**

Typical molding:
Aligned skin, randomized core



Injection speed: **2.5 in/sec**

a_{11}	a_{22}	a_{33}
0.90 ± 0.02	0.06 ± 0.02	0.04 ± 0.02



Injection speed: **4 in/sec**

Macro to Micro Properties: **Crystallinity**

Watch for **lower crystallinity** due to faster cooling (quenching)

Mechanical strength can vary dramatically ***despite parts filling***

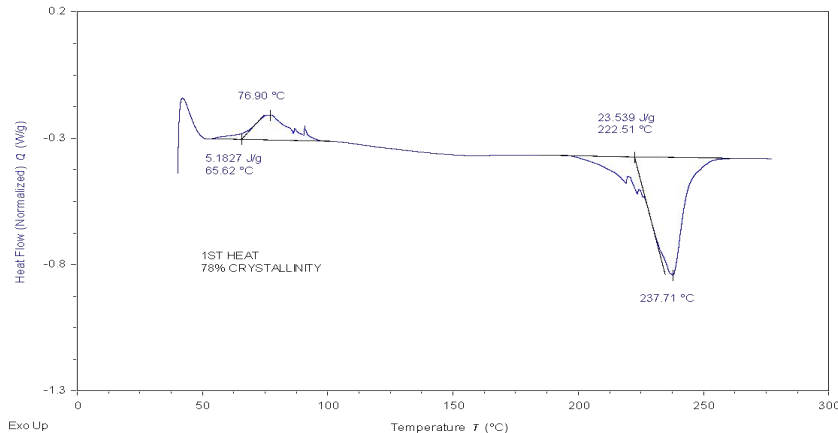


Psylotech tensile specimens
(Ixef 1022)

Mold temp: **310°F**

Crystallinity = 78%

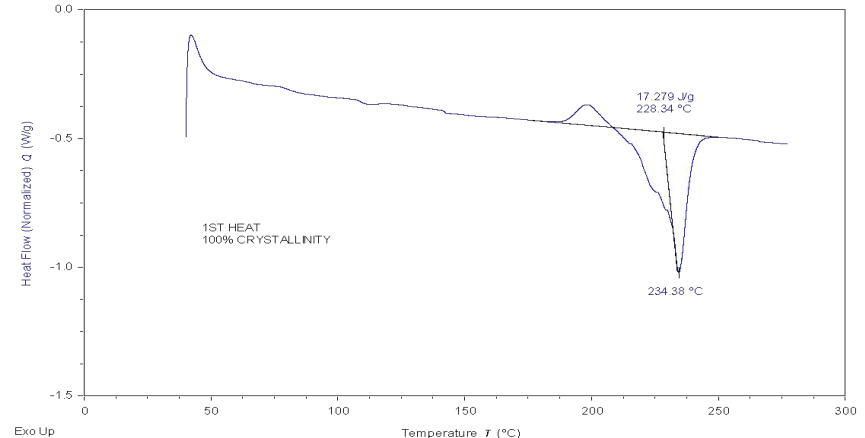
Tensile Strength = **80±8.7 MPa** (Average of 5 tests)



Mold temp: **320°F**

Crystallinity = 100%

Tensile Strength = **104±4 MPa** (Average of 5 tests)



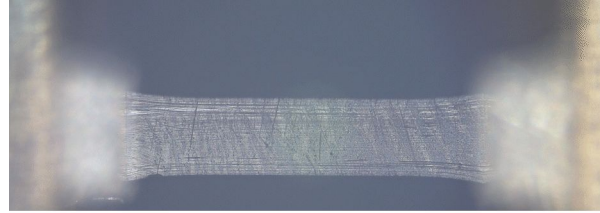
Macro to Micro Properties: Mechanical Strength (Method)

Psylotech tensile bars are pulled on a load frame in our polymer physics lab.

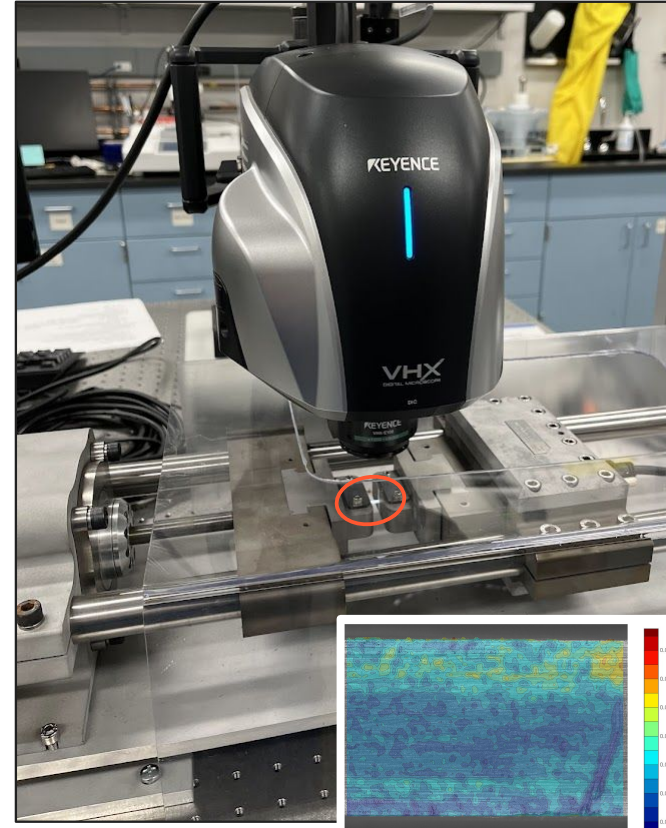
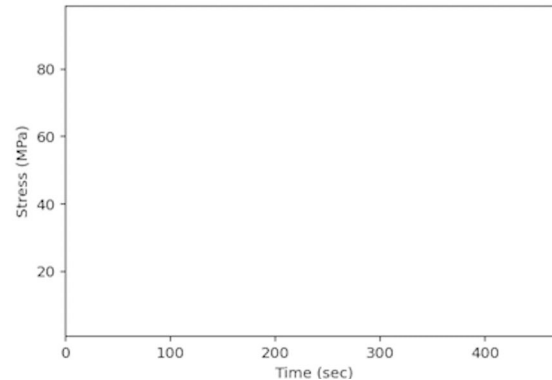
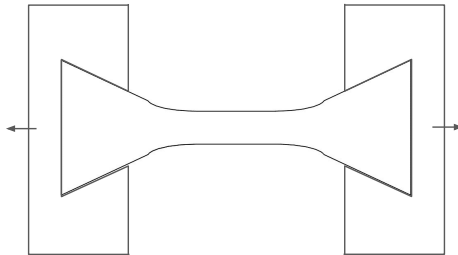
The frame is on an air table to eliminate vibration, with an optical microscope mounted above the testing area.

Force is measured by a load cell

Strain is calculated by DIC (Digital Image Correlation), eliminating error associated with any fixturing slip.



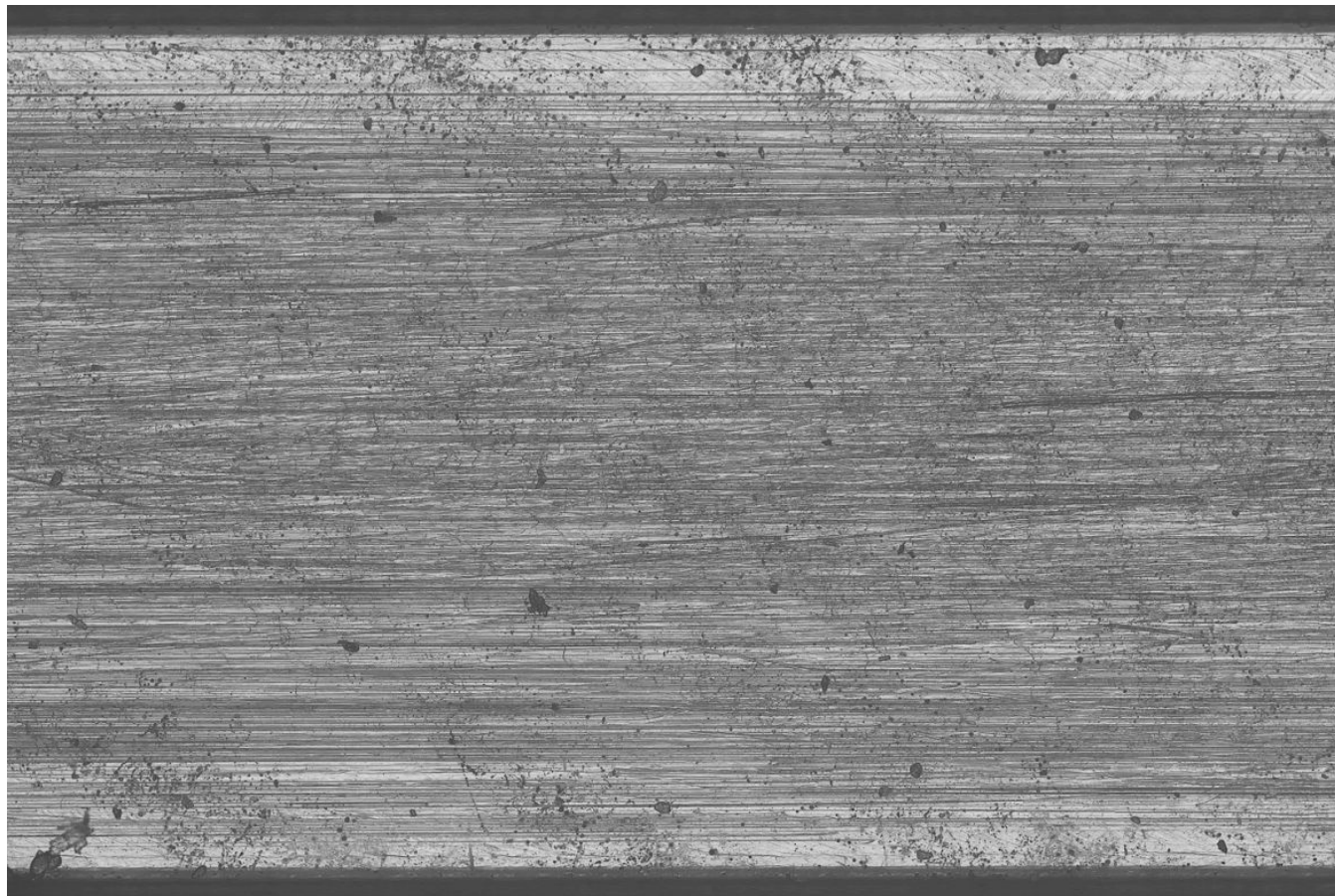
Ductile failure of Radel R-5000



Macro to Micro Properties: **Mechanical Strength**

Preliminary results show **equivalent or improved mechanical performance.**

Higher modulus and force at break can be attributed to fiber alignment and/or size effects.



Surface defects at high stresses during ductile failure of Radel® R-5000.

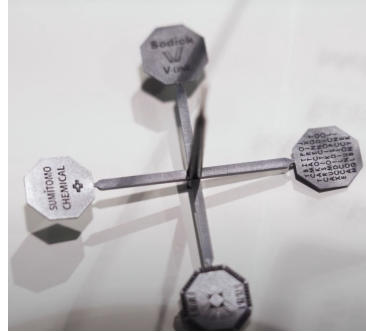
Macro to Micro Properties: **Moisture Uptake Rate**

Absorption is a function of surface area

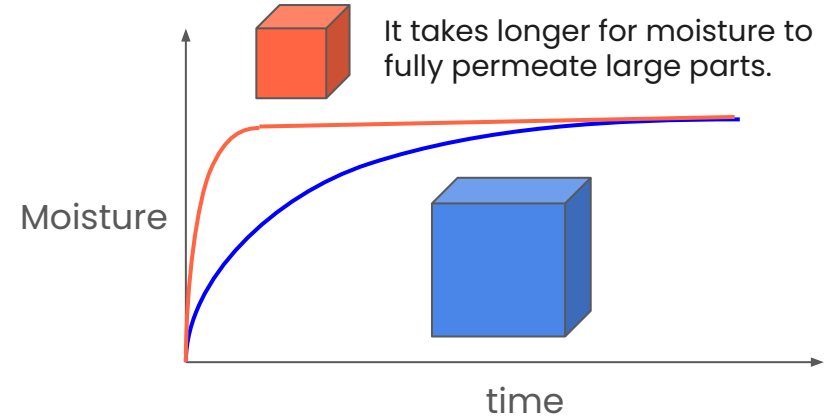
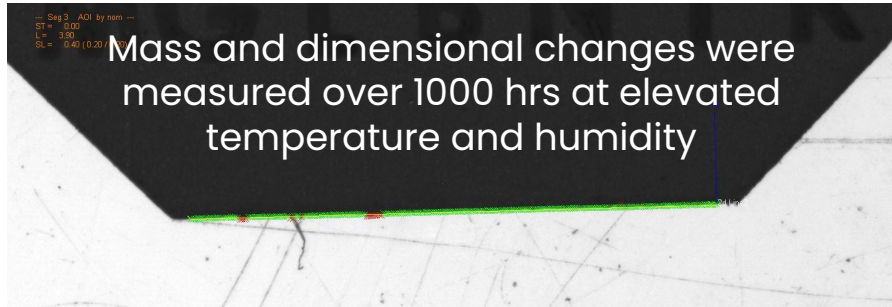
Moisture capacity is a function of volume

Micro parts have a higher **surface area / volume ratio**

Takeaway: Moisture saturation values are generally the same as macro, but they are reached faster

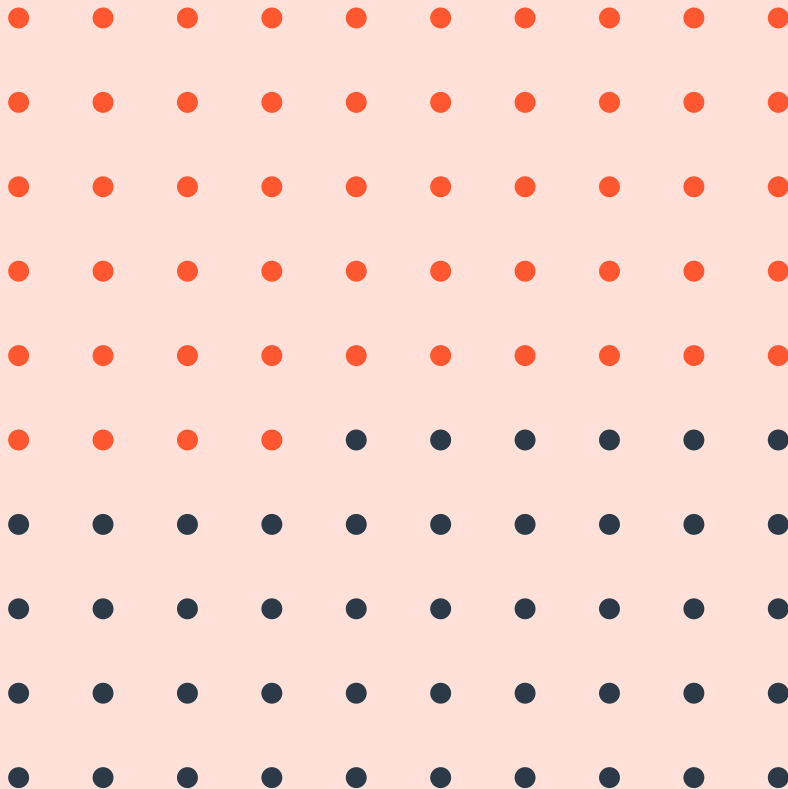


(Placeholder,
add graph with
1000-hr data)





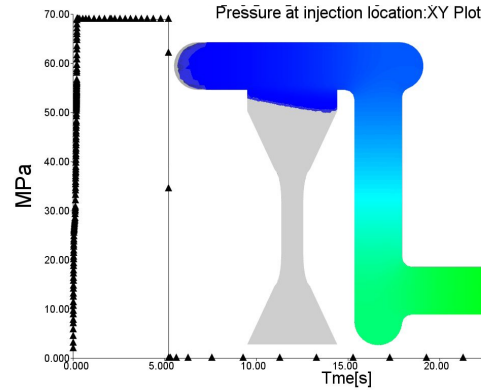
Micromolding Simulation



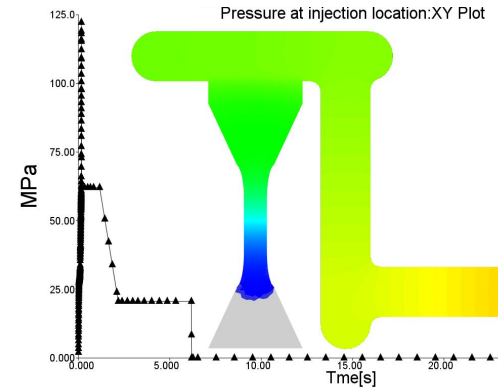
Macro to Micro: Simulation

Scientific molding principles (97% V/P switchover) should NOT be utilized when the part is small relative to the filling system.

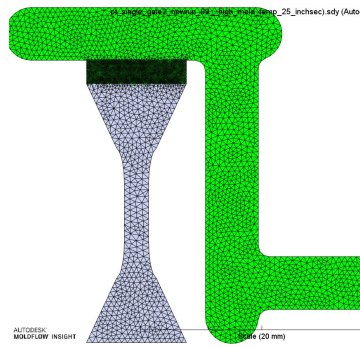
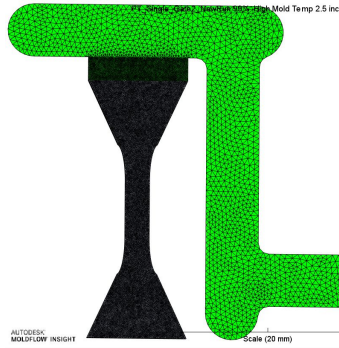
SIMULATE WITH FILL ONLY. NO PACKING



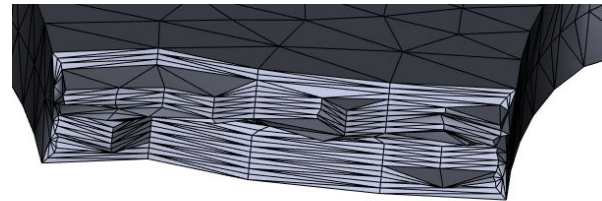
97% = 70 MPa



99% = 125 MPa



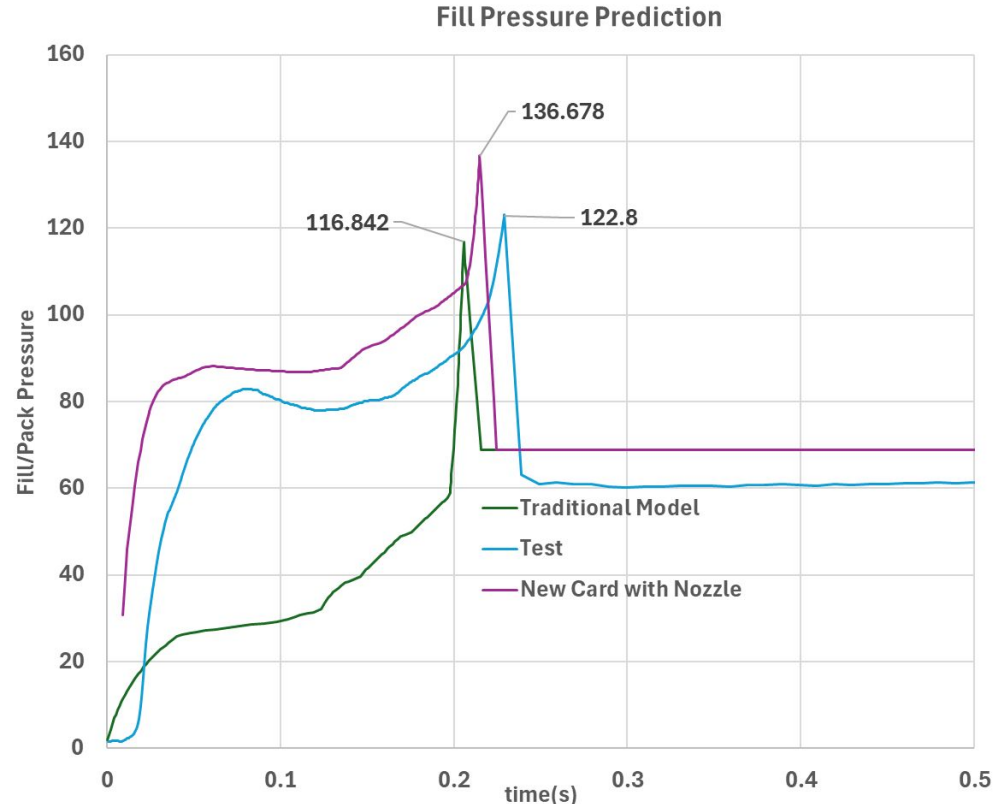
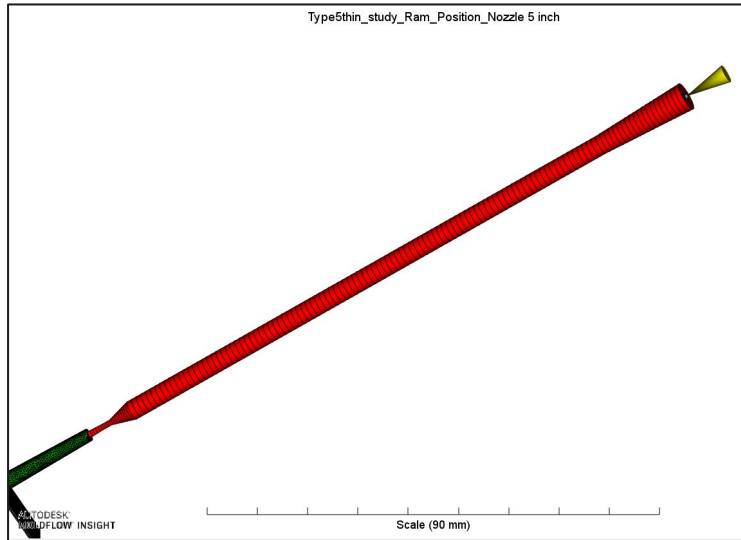
The simulation **resolution** (mesh element size, and the number of layers across thickness) was not impactful in our case.



No change between 10 and 20 layers across thickness

Macro to Micro: Simulation

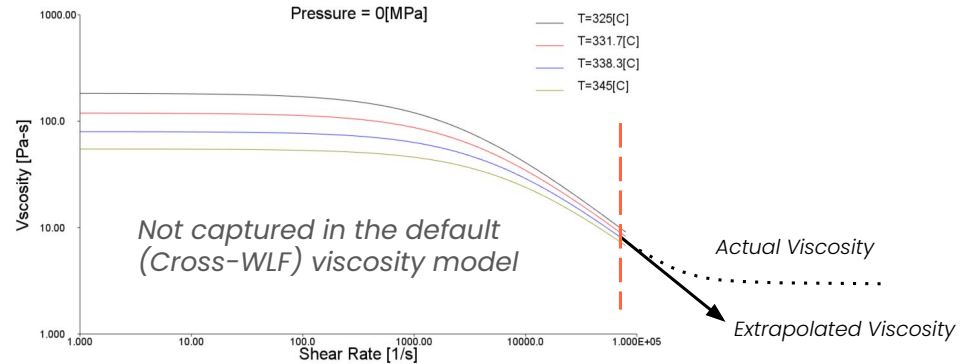
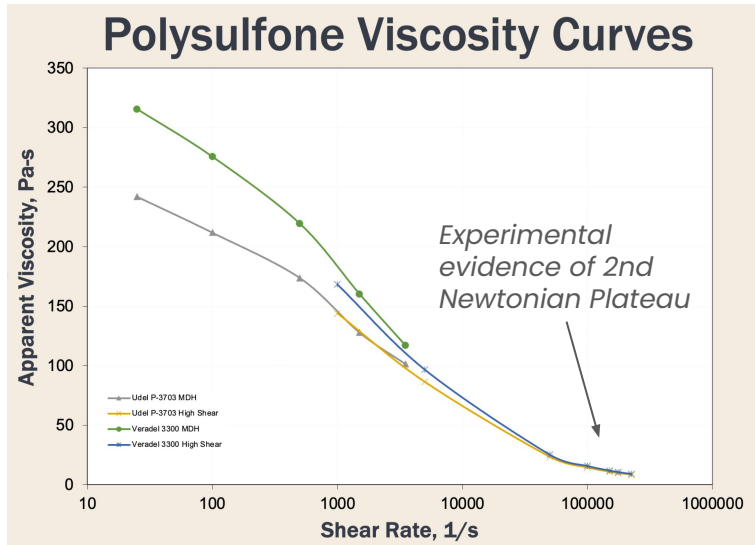
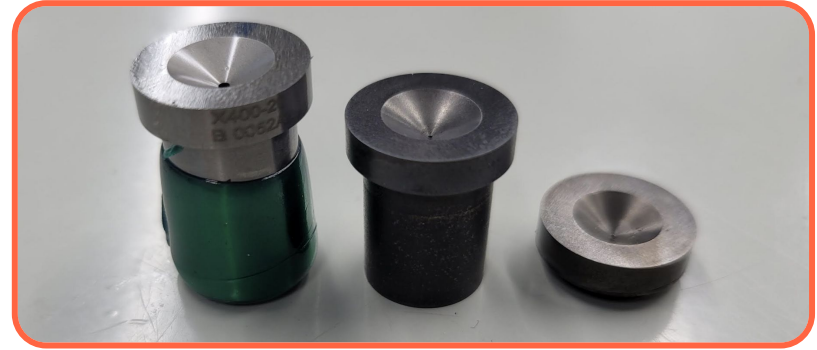
Modelling the nozzle flow constriction (as a hot runner) **raised the pressure by 17%** and resulted in a more conservative prediction result.



Macro to Micro: Simulation

High shear rheology data challenges:

- Maxing out the load cell
- Cleaning the micro-diameter dies
- Outsourcing unsuccessful so far



Current efforts: using simulation/experimental divergence to reverse engineer viscosity data.

Macro to Micro: **Takeaways**

Findings:

-
-

We are now capable of characterization of material for micromolding applications.
Contact us for material evaluation & de-risking for your application.

QR code to these slides
(PDF)

References, appendix (∞ slides)

Parts to bring:

- Zeniva ZA-600 tensile bars
- Mini-discs and rods
- Spiral flow parts
- Matrix tool